

Adham Ashraf Saeed

Assignment 1 (SQL)

Access the following Database

```
1 CREATE DATABASE sql_practice;
2
3 USE sql_practice;
4
5 CREATE TABLE orders (
6     orderid INT PRIMARY KEY,
7     customerid VARCHAR (20) NOT NULL,
8     productname VARCHAR (100) NOT NULL,
9     category VARCHAR (50) NOT NULL,
10    country VARCHAR (50) NOT NULL,
11    orderdate DATE NOT NULL,
12    shipdate DATE NULL,
13    quantity INT NOT NULL,
14    unitprice DECIMAL (10, 2) NOT NULL,
15    sales DECIMAL (10, 2) NOT NULL,
16    discount DECIMAL (10, 2) NULL,
17    profit DECIMAL (10, 2) NOT NULL
18 );
19
20 INSERT INTO orders (orderid, customerid, productname, category, country, orderdate, shipdate, quantity, unitprice, sales, discount, profit)
21 VALUES
22 (1, 'c101', 'office chair', 'furniture', 'usa', '2023-01-15', '2023-01-18', 2, 250.00, 500.00, 0.10, 80.00),
23 (2, 'c102', 'gaming chair', 'furniture', 'usa', '2023-03-10', '2023-03-12', 1, 700.00, 700.00, 0.20, 120.00),
24 (3, 'a103', 'laptop', 'technology', 'germany', '2023-05-05', '2023-05-08', 3, 400.00, 1200.00, 0.15, 250.00),
25 (4, 'b104', 'desk lamp', 'office supplies', 'uk', '2022-11-20', '2022-11-25', 5, 40.00, 200.00, 0.05, 30.00),
26 (5, 'c105', 'tablet', 'technology', 'usa', '2023-07-01', NULL, 1, 350.00, 350.00, NULL, -20.00),
27 (6, 'd106', 'bookshelf', 'furniture', 'canada', '2024-01-12', '2024-01-15', 4, 150.00, 600.00, 0.25, 90.00),
28 (7, 'c107', 'monitor', 'technology', 'usa', '2023-09-09', '2023-09-14', 2, 275.00, 550.00, 0.00, 60.00),
29 (8, 'e108', 'printer', 'technology', 'france', '2023-04-18', NULL, 1, 180.00, 180.00, 0.10, -15.00),
30 (9, 'c109', 'wooden chair', 'furniture', 'uk', '2023-12-05', '2023-12-09', 6, 120.00, 720.00, 0.30, 140.00),
31 (10, 'f110', 'mouse', 'technology', 'germany', '2023-06-22', '2023-06-24', 10, 25.00, 250.00, 0.05, 40.00),
32 (11, 'c111', 'conference chair', 'furniture', 'usa', '2022-08-14', '2022-08-19', 3, 320.00, 960.00, 0.10, -50.00),
33 (12, 'g112', 'phone', 'technology', 'india', '2023-02-11', '2023-02-15', 2, 500.00, 1000.00, 0.20, 200.00),
34 (13, 'h113', 'binder', 'office supplies', 'usa', '2023-10-01', '2023-10-03', 1, 15.00, 15.00, 0.00, 5.00),
35 (14, 'c114', 'standing desk', 'furniture', 'germany', '2023-11-19', '2023-11-22', 2, 650.00, 1300.00, 0.10, 300.00),
36 (15, 'i115', 'headphones', 'technology', 'usa', '2024-02-05', '2024-02-07', 4, 80.00, 320.00, 0.05, 45.00),
37 (16, 'c116', 'study chair', 'furniture', 'usa', '2023-08-08', NULL, 5, 210.00, 1050.00, 0.40, 180.00),
38 (17, 'j117', 'router', 'technology', 'uk', '2023-03-30', '2023-04-02', 2, 150.00, 300.00, NULL, -10.00),
39 (18, 'c118', 'filing cabinet', 'furniture', 'canada', '2023-05-21', '2023-05-25', 1, 450.00, 450.00, 0.20, 70.00),
40 (19, 'k119', 'desk chair pro', 'furniture', 'usa', '2023-06-01', '2023-06-05', 2, 480.00, 960.00, 0.10, 150.00),
41 (20, 'c120', 'server rack', 'technology', 'germany', '2023-07-17', '2023-07-20', 1, 1100.00, 1100.00, 0.05, 220.00);
```


Basic Questions

1. Write a query to display all columns from the Orders table.

```
SELECT *  
FROM orders;
```

2. Display only the OrderID, CustomerID, and Sales columns.

```
SELECT orderid,  
       customerid,  
       sales  
FROM orders;
```

3. Show all orders where Sales > 500.

```
SELECT *  
FROM orders  
WHERE sales > 500;
```

4. Retrieve orders where Quantity = 1.

```
SELECT *  
FROM orders  
WHERE quantity = 1;
```

5. Display all orders where Country = 'USA'.

```
SELECT *  
FROM orders  
WHERE country = 'USA';
```


Basic Questions

6. List all distinct countries from the Orders table.

```
SELECT DISTINCT country  
FROM orders;
```

7. Show orders where Profit < 0.

```
SELECT *  
FROM orders  
WHERE profit < 0;
```

8. Retrieve orders where Discount >= 0.2.

```
SELECT *  
FROM orders  
WHERE discount >= 0.2;
```

9. Display orders where ShipDate IS NULL.

```
SELECT *  
FROM orders  
WHERE shipdate IS NULL;
```

10. Show orders where Sales = Quantity * UnitPrice

```
SELECT *  
FROM orders  
WHERE sales = quantity * unitprice;
```


Intermediate Questions

1. Retrieve orders where Country = 'USA' AND Sales > 300.

```
SELECT *  
FROM orders  
WHERE country = 'USA'  
AND sales > 300;
```

2. Show orders where Category = 'Furniture' OR Category = 'Technology'.

```
SELECT *  
FROM orders  
WHERE category = 'Furniture'  
OR category = 'Technology';
```

3. Find all orders where CustomerID starts with 'C' using LIKE.

```
SELECT *  
FROM orders  
WHERE customerid LIKE 'C%';
```

4. Display orders where OrderDate is BETWEEN '2023-01-01' AND '2023-12-31'.

```
SELECT *  
FROM orders  
WHERE orderdate BETWEEN '2023-01-01' AND '2023-12-31';
```

5. Retrieve orders where Country IN ('USA', 'UK', 'Germany'), ordered by Sales DESC.

```
SELECT *  
FROM orders  
WHERE country IN ('USA', 'UK', 'Germany')  
ORDER BY sales DESC;
```


Advanced Questions

1. Show orders where Sales > 500 AND Profit > 0, ordered by Profit DESC.

```
SELECT *
FROM orders
WHERE sales > 500
      AND profit > 0
ORDER BY profit DESC;
```

2. Retrieve orders where Discount IS NOT NULL AND Discount > 0, sorted by Discount ASC.

```
SELECT *
FROM orders
WHERE discount IS NOT NULL
      AND discount > 0
ORDER BY discount ASC;
```

3. Find orders where (Quantity * UnitPrice) - Discount > 1000.

```
SELECT *
FROM orders
WHERE (quantity * unitprice) - discount > 1000;
```

4. Display orders where Product name contains 'Chair', and Sales BETWEEN 200 AND 1000.

```
SELECT *
FROM orders
WHERE productname LIKE '%Chair%'
      AND sales BETWEEN 200 AND 1000;
```

5. Retrieve distinct CustomerID values where Country = 'USA' AND Profit < 0, ordered alphabetically.

```
SELECT DISTINCT customerid
FROM orders
WHERE country = 'USA'
      AND profit < 0
ORDER BY customerid ASC;
```